



SO-101 Calibration Session

CS 6341 Robotics

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LeRobot SO-101 Robot



so101_leader



so101_follower

Preparation

- One member of each group come get your SO101 robot.
- Check that everything is included in the bags.
- Fill up and the Sign one copy of the Material List and give it to the TA.
- Organize your working space.

Target

- **Set up the LeRobot SO-101 environment.**
- **Complete the follower robot calibration.**



Robot Configuration

- Follow the instructions to setup an Anaconda environment for the LeRobot libraries.

<https://huggingface.co/docs/lerobot/installation>

- Follow the instructions to setup to configure and calibrate robots.

<https://huggingface.co/docs/lerobot/so101>



Environment Setup

Create the conda environment

```
conda create -y -n lerobot  
python=3.12  
conda activate lerobot
```

Install PyTorch

```
(support PyTorch >= 2.10)  
pip install torch torchvision
```

Clone the LeRobot repository

```
git clone  
https://github.com/huggingface/lerobot.git  
cd lerobot
```

Install LeRobot for SO-101

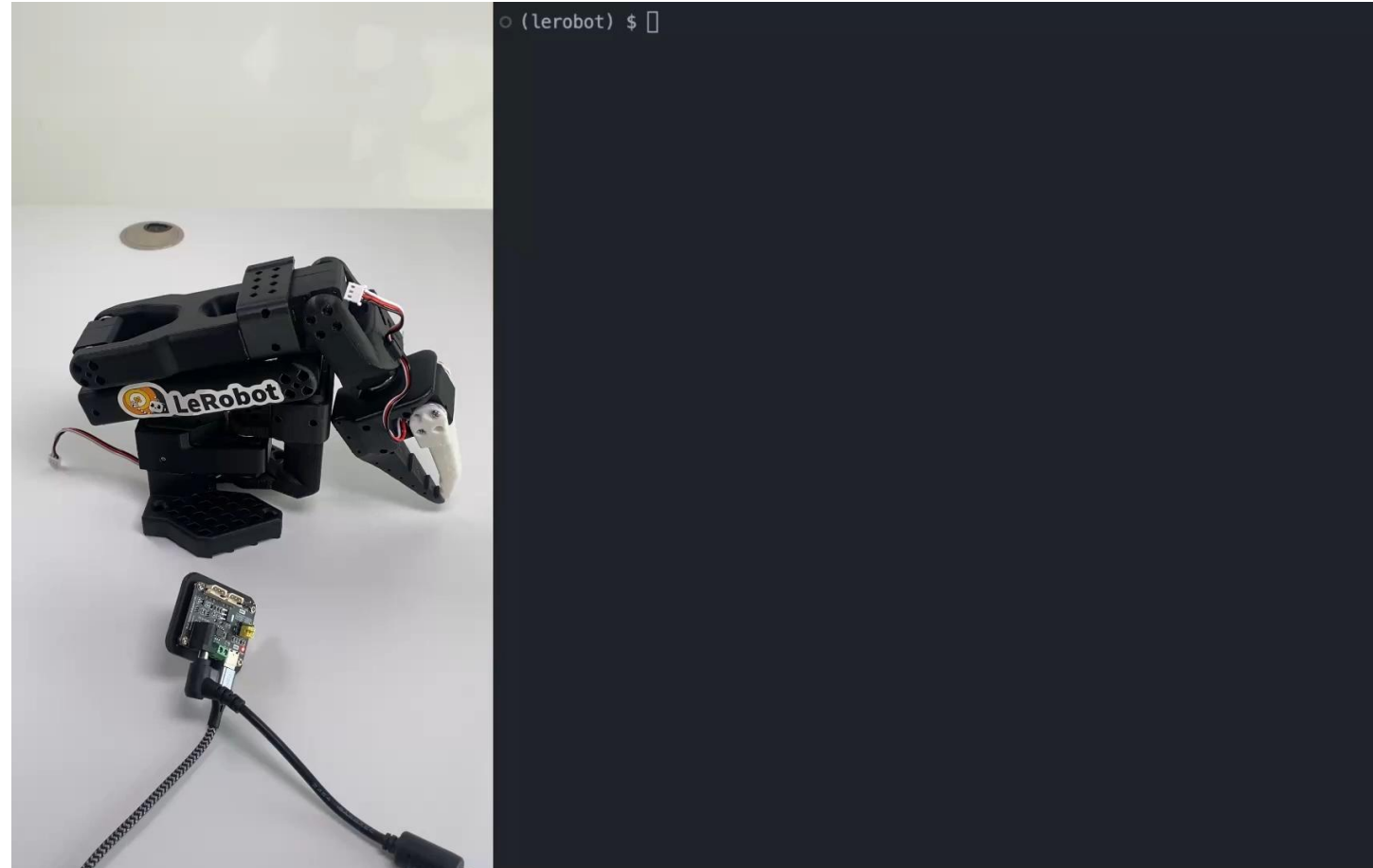
```
pip install -e ".[core_scripts,feetech]"
```



<https://huggingface.co/docs/lerobot/installation>

Bus Adapter

- Connect the bus adapter to the PC using a cable.
- Connect the adapter to the power supply.



Find the USB ports associated with each arm

To find the port for each bus adapter, connect MotorBus to your computer via USB and power. Run the following script and disconnect the MotorBus when prompted:

```
lerobot-find-port
```

Example output:

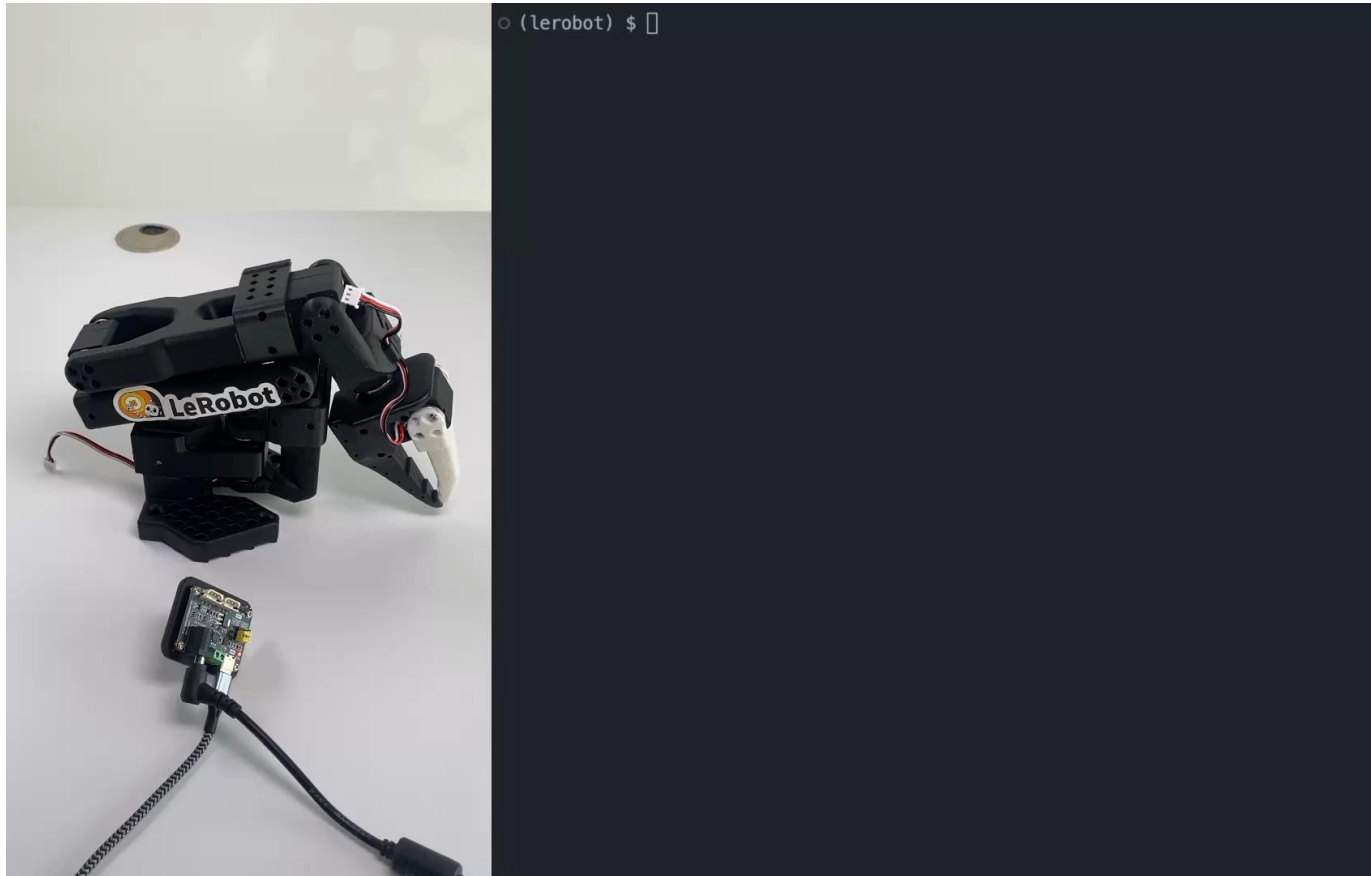
```
(lerobot) C:\Users\zqfan\lerobot_workspace\lerobot>lerobot-find-port
Finding all available ports for the MotorsBus.
Ports before disconnecting: ['COM3']
Remove the USB cable from your MotorsBus and press Enter when done.
|
```

```
(lerobot) C:\Users\zqfan\lerobot_workspace\lerobot>lerobot-find-port
Finding all available ports for the MotorsBus.
Ports before disconnecting: ['COM3']
Remove the USB cable from your MotorsBus and press Enter when done.

The port of this MotorsBus is 'COM3'
Reconnect the USB cable.
```

Setup motors

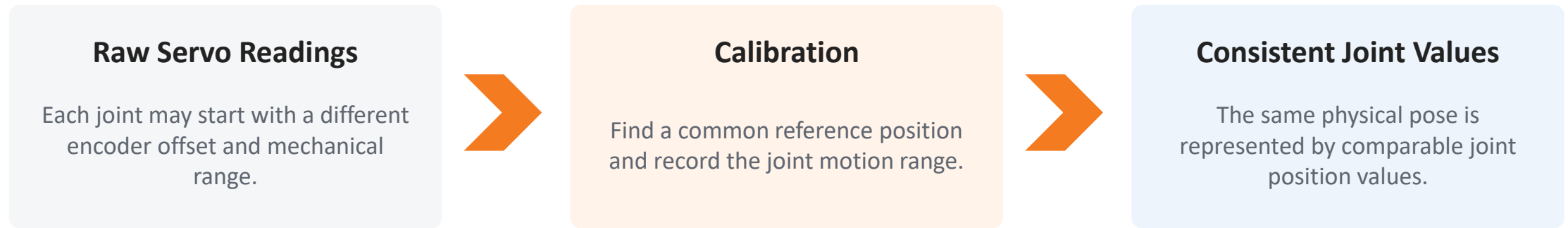
Each motor must have a unique ID.



- **The Motor ID setup has already been completed**, so you do not need to run it again.
- Remember the motor order from top to bottom: **Gripper → Wrist Roll → Wrist Flex → Elbow Flex → Shoulder Lift → Shoulder Pan**
- If a motor is replaced, run the Motor ID setup again for the replaced motor.

SO-101 Calibration: Why?

Calibration aligns the robot's position reference and usable joint range.



Key idea

Calibration makes leader and follower position values correspond to the same physical joint positions, which is important for teleoperation and transferring learned behaviors between robots.

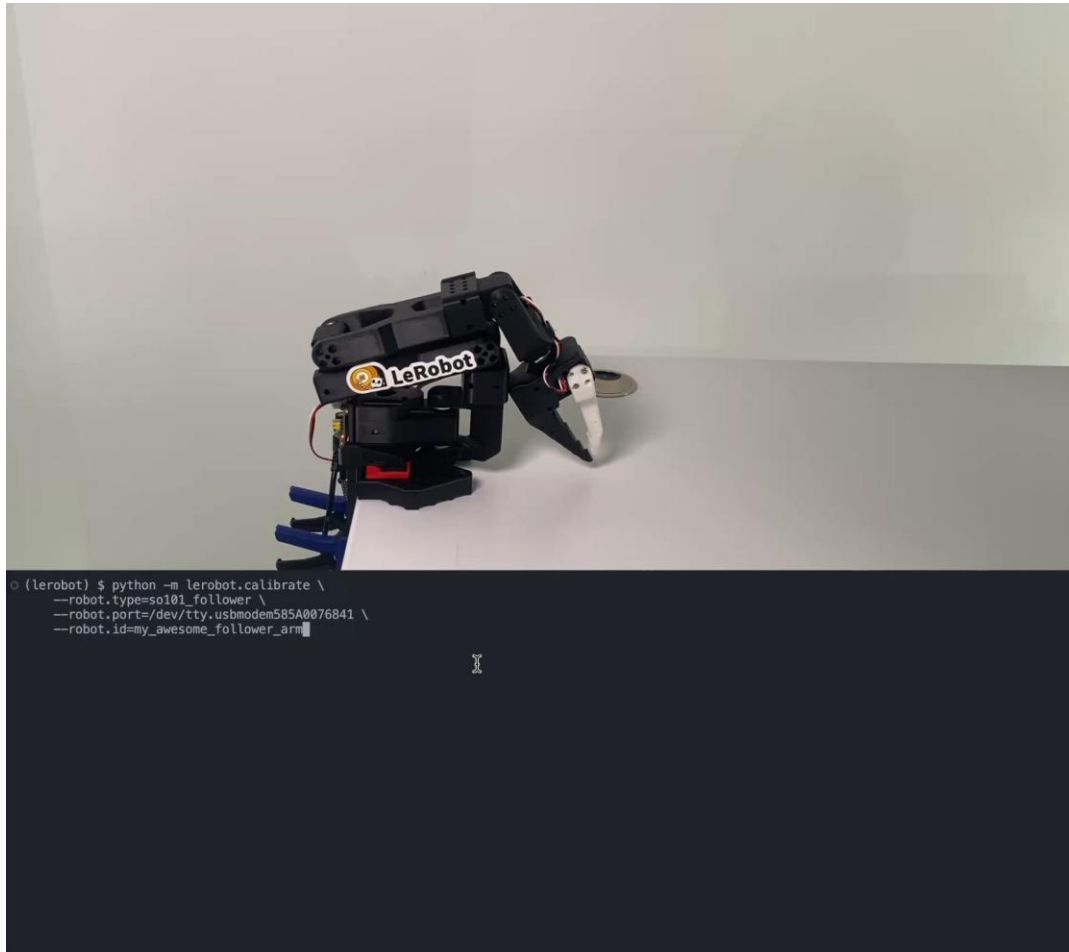
What Happens During Calibration?



Calibration changes position reference and joint range.

Calibrate the follower arm:

```
lerobot-calibrate \  
  --robot.type=so101_follower \  
  --robot.port=/dev/tty.usbmodem58760431551 \ # <- The port of your robot  
  --robot.id=my_awesome_follower_arm # <- Give the robot a unique name
```



- You only need to complete the **follower arm calibration**. The leader arm has already been calibrated.
- After calibration, send the saved **calibration file (.json)** to my email: Qifan.Zhang@utdallas.edu. Then come to me for a teleoperation test.

Calibration saved to C:\Users\zqfan\.cache\huggingface\lerobot\calibration\robots\so_follower\my_awesome_follower_arm.json